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TOPICAL REVIEW

# From Gums to Brain: Unveiling the Hidden Inflammatory Connection Between Periodontal and Cerebrovascular Diseases

Jussie C. Lima, MD, DDS*; Mohamed F. Doheim, MD*; Camille C.S. Nogueira, DDS; Souvik Sen, MD; Raul G. Nogueira, MD

**ABSTRACT:** Periodontal disease (PD) encompasses inflammatory conditions affecting the supporting structures of the teeth, with gingivitis and periodontitis being the most prevalent forms. Gingivitis, characterized by gum inflammation and bleeding, can progress to periodontitis, which involves the destruction of connective tissue and alveolar bone. PD is a global health burden, affecting over a billion individuals, with 42.2% of adults aged ≥30 years in the United States experiencing some form of periodontitis. Without proper management, PD can lead to severe systemic consequences. The cause of PD is multifactorial, involving bacterial biofilm, systemic diseases, aging, smoking, high-fat diets, and immune deficiencies. These risk factors act independently or synergistically, accelerating disease progression. Prevention and treatment strategies focus on biofilm control, nonsurgical therapy, improved oral hygiene, and risk factor management to halt disease progression and restore tooth support. Recent research suggests a strong link between PD and cardiovascular diseases, including stroke, due to systemic inflammation. Periodontitis-induced inflammation triggers the release of cytokines and CRP (C-reactive protein), which can enter systemic circulation, promoting atherosclerosis and endothelial dysfunction. Bacterial translocation from the oral cavity into the bloodstream further exacerbates vascular disease, increasing stroke risk. Epidemiological studies indicate that individuals with periodontitis have a significantly higher risk of stroke, with case-control and cohort studies showing strong associations. However, some interventional studies suggest that periodontal treatment alone may not significantly reduce cardiovascular events, highlighting the complexity of this relationship. This review aimed to summarize existing literature on the association between PD and cerebrovascular diseases, emphasizing underlying pathophysiological mechanisms and epidemiological evidence.

**GRAPHIC ABSTRACT:** A graphic abstract is available for this article.

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**Key Words:** cardiovascular diseases ■ C-reactive protein ■ dental plaque ■ dysbiosis ■ fibrinolysis

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Periodontal diseases (PDs) are inflammatory conditions, with gingivitis being the most common type, characterized by gum inflammation and bleeding.[1] It represents a significant public health issue in the United States, encompassing both gingivitis and periodontitis.[1,2] According to the recent data, ≈42.2% of adults aged ≥30 years are affected by some form of periodontitis, with 7.8% experiencing severe cases and 34.4% having nonsevere forms. PD has a high prevalence worldwide, where more than a billion (19%) of the world population have severe PD.[3,4] Without proper treatment, gingivitis can progress to periodontitis, a more severe condition marked by the destruction of the connective tissue attached to the gums and the loss of alveolar bone.[1,2,4,5]

Various factors contribute to the development of PD, including bacterial presence, systemic diseases, smoking, aging, high-fat diets, and autoimmune diseases.[6] These factors may act independently or synergistically, exacerbating disease progression. Effective prevention and treatment strategies focus on controlling bacterial biofilm, including nonsurgical therapy, combined with improving personal oral hygiene, managing risk factors,

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halting disease progression, and restoring lost tooth support.[7,8]

The inflammatory process plays a critical role in atherosclerosis pathogenesis.[9] Numerous studies have explored the relationship between periodontitis, systemic inflammation, and cardiovascular disease. Periodontitis is primarily driven by microbial dysbiosis, mainly by the red complex represented by *Porphyromonas gingivalis*, *Tannerella forsythia*, and *Treponema denticola*, which play a central role in this process. These bacteria possess virulence mechanisms that disrupt host-microbe hemostasis and initiate a strong inflammatory immune response.[10,11]

Dental plaque and biofilms trigger host response to combat bacterial invasion, which may inadvertently lead to the destruction of supporting connective tissue.[12] Inflammation originating from periodontitis can disseminate systemically, increasing inflammatory markers, such as CRP (C-reactive protein) and white blood cell counts, having been observed in patients with periodontitis compared with those without.[13] It is established that CRP levels have been independently linked to cardiovascular disease events.[14] Chronic systemic inflammatory activation may contribute to atherosclerosis through multiple pathways, influencing atherogenesis, coagulation, and fibrinolysis, ultimately elevating the risk of adverse cardiovascular events.[15]

Cerebrovascular diseases are prevalent, have high rates of morbidity and mortality, and are on the rise globally.[16] Despite a decline in the stroke death rate from 41.1 per 100 000 in 2021 to 39.5 per 100 000 in 2022, stroke remains one of the leading causes of death in the United States.[17] Approximately 795 000 individuals experience a new or recurrent stroke each year, with about 610 000 first attacks and 185 000 recurrent events. Ischemic stroke accounts for 87% of cases, while intracerebral hemorrhage and subarachnoid hemorrhage account for 10% and 3%, respectively.[18] Large-artery atherosclerotic disease strokes resulting from intracranial atherosclerotic disease carry a poor prognosis, marked by worse functional outcome increase morbidity and mortality compared with the other stroke subtypes.[19]

Periodontitis may indirectly contribute to stroke through inflammatory processes, including bacterial translocation, cytokine production, and systemic inflammation, which promote atherosclerosis and increase stroke risk.[20–23] This review aimed to provide a comprehensive synthesis of current evidence on the interplay between PD and cerebrovascular disorders, organized across 3 dimensions: mechanistic pathways, epidemiological associations, and clinical implications. Our objective was not only to summarize existing research but also to critically appraise the strength of causal inference and explore potential preventive strategies. We followed a structured progression beginning with pathophysiological mechanisms, moving through observational and interventional evidence, and concluding with translational considerations for clinical practice.

## PATHOPHYSIOLOGICAL MECHANISMS

The connection between periodontitis and stroke is largely mediated by the inflammatory processes that occur in both conditions (Figure 1). Periodontitis, a chronic inflammatory disease affecting the gums and supporting structures of the teeth, leads to the release of proinflammatory factors such as cytokines and prostaglandins.[6] These inflammatory markers can enter the bloodstream and contribute to systemic inflammation, potentially promoting atherosclerosis and endothelial dysfunction.[20–23]

The equilibrium of the oral microbiota is essential for maintaining host health, homeostasis, and immunity. This complex microbial community comprises ≈700 species classified into 185 genera and 12 phyla, including *Firmicutes*, *Fusobacteria*, *Actinobacteria*, *Proteobacteria*, *Bacteroidetes*, *Chlamydiae*, *Chloroflexi*, and *Spirochaetes*. The interactions between these microorganisms and their hosts are critical for maintaining oral health in commensal, symbiotic, and pathogenic forms.[24,25] A reduction in beneficial bacteria coupled with an increase in pathogenic species disrupts this balance, leading to oral dysbiosis. This shift can result from factors such as poor oral hygiene, diet, antibiotic use, and underlying health conditions.[24,25] Consequently, dysbiosis facilitates the overgrowth of opportunistic pathogens such as *P gingivalis*, *T forsythia*, *T denticola*, *Aggregatibacter actinomycetemcomitans*, *Prevotella intermedia*, *Fusobacterium nucleatum*, *Eikenella corrodens*, and *Candida albicans*, which are implicated in diseases including dental caries, PD, and oral candidiasis. The red complex bacteria (*P gingivalis*, *T forsythia*, and *T denticola*) are particularly significant in PD due to their virulence, ability to subvert immune responses, and synergistic interactions within the biofilm. Furthermore, this microbial dysbiosis is not confined to local infection; oral pathogens can spread from the subgingival biofilm into the bloodstream, allowing them to disseminate to various organs and tissues. This dissemination may play a pivotal role in the development of multiple systemic diseases by contributing to inflammation and pathogenic processes beyond the oral cavity.[24,25]

Given the rich vascular supply to the oral cavity, including the lingual, maxillary, and facial arteries, which branch from the external carotid artery, the infection and inflammation in the gingiva can gain direct access to the systemic circulation, thereby affecting the cerebral vasculature. As a result, periodontitis may accelerate the processes leading to cerebrovascular events such as stroke, especially in individuals with other risk factors for cardiovascular disease.[20–23] Chronic inflammatory diseases are multifactorial conditions that can lead to significant tissue damage and contribute to a wide range of systemic

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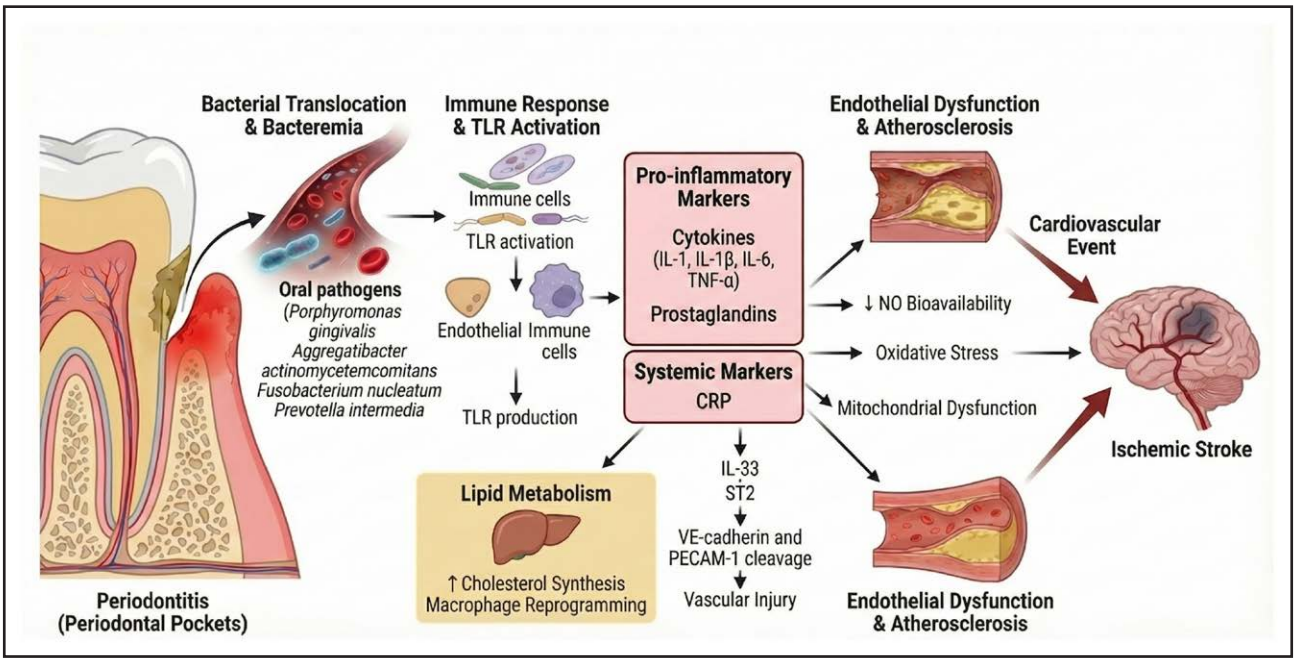


**Figure 1. Pathophysiological pathway linking periodontitis to ischemic stroke.**
Bacterial translocation of periodontal pathogens (eg, *P gingivalis* and *P intermedia*) from inflamed pockets triggers an immune response via TLR (toll-like receptor) activation. This initiates a systemic inflammatory cascade characterized by elevated cytokines (IL [interleukin]-1, IL-6, and TNF-α [tumor necrosis factor-α]) and CRP (C-reactive protein). Subsequent dysregulation of lipid metabolism, oxidative stress, mitochondrial dysfunction, and the IL-33/ST2 (suppression of tumorigenicity 2) axis promotes endothelial dysfunction and atherosclerosis, culminating in an increased risk of ischemic stroke. PECAM-1 indicates platelet endothelial cell adhesion molecule-1; and VE-cadherin, vascular endothelial cadherin.

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health complications. The pathogenesis of these diseases involves a complex interplay of genetic, environmental, and immune system factors, resulting in diverse host responses that can vary considerably between individuals. Genetic predisposition, environmental exposures, and immune dysregulation are all critical factors in the onset and progression of chronic inflammation, often leading to unpredictable outcomes.[26,27] These conditions have been associated with a 28% increased risk of cardiovascular disease.[28] Persistent systemic inflammation, such as that induced by periodontitis, may promote endothelial dysfunction through a variety of mechanisms, further exacerbating the risk of cardiovascular complications (Figure 1).[29]

Bacterial translocation from the oral cavity into the bloodstream plays a pivotal role in the development of atherosclerosis, particularly in individuals with chronic PD. Studies following scaling procedures have consistently demonstrated significantly higher levels of bacteremia in patients with periodontitis compared with those with gingivitis or healthy controls. In these patients, the extent of bacteremia was strongly associated with clinical parameters such as the gingival index, plaque index, and the number of sites with bleeding on probing, but not with probing pocket depth.[29,30] While the immune system typically clears blood-borne bacteria within minutes, conditions such as infective endocarditis can promote bacterial colonization at specific vascular sites.[31] Clinical studies have confirmed the presence of periodontal pathogens in atherosclerotic plaques, with an analysis of carotid atherosclerotic samples from 37 patients revealing the predominant isolation of *P gingivalis* and *P intermedia* (Gram-negative rods) and *Propionibacterium acnes* (Gram-positive rods).[32] Research also indicates that the coronary artery is most frequently affected, followed by the carotid, abdominal, mammary, and femoral arteries (Figure 1).[33]

Recent mechanistic studies have clarified the molecular cascades linking oral pathogens to vascular inflammation and endothelial injury. *P gingivalis*, the principal periodontal bacterium, activates TLRs (toll-like receptors) 2 and 4 on endothelial and immune cells, initiating MyD88-dependent signaling and NF-κB (nuclear factor-κB) activation.[20] Nuclear translocation of NF-κB induces expression of IL (interleukin)-6, TNF-α (tumor necrosis factor-α), and IL-1β, stimulating hepatic production of CRP and amplifying systemic inflammation.[34] Elevated CRP reduces NO bioavailability, upregulates endothelial adhesion molecules (ICAM-1 [intercellular adhesion molecule-1], VCAM-1 [vascular cell adhesion molecule-1], and E-selectin), and promotes monocyte adhesion.[20,35] In vitro and animal studies demonstrate that chronic *P gingivalis* exposure accelerates atherogenesis in ApoE-deficient mice and disrupts endothelial junctions through gingipain-mediated cleavage of VE-cadherin (vascular endothelial cadherin) and PECAM-1 (platelet

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endothelial cell adhesion molecule-1), while bacterial outer membrane vesicles further increase vascular permeability.[36,37] Other periodontal pathogens exert distinct vascular effects: *A actinomycetemcomitans* LtxA (leukotoxin A) upregulates ICAM-1 and VCAM-1, inducing a proinflammatory phenotype in human brain endothelial cells, whereas *Fusobacterium nucleatum* preferentially triggers TLR4-MyD88-NF-κB signaling, underscoring species-specific endothelial activation.[38–41] Clinically, patients with periodontitis show elevated IL-6 and CRP levels together with impaired flow-mediated vasodilation, linking the TLR→NF-κB→IL-6/CRP pathway to functional vascular injury and heightened stroke risk.[42,43]

Beyond these immunologic cascades, *P gingivalis* directly infects vascular endothelial cells, inducing mitochondrial dysfunction, oxidative stress, and impaired endothelial integrity.[37,43] These changes lead to endothelial cell apoptosis, endothelial-to-mesenchymal transition, and reduced repair capacity. *P gingivalis* also increases endothelial permeability by proteolytically cleaving adhesion molecules such as PECAM-1 and VE-cadherin while reducing plasminogen activator inhibitor-1, further exacerbating dysfunction. The pathogen's effects on nitric-oxide homeostasis, through decreased NO production and enhanced inactivation, further compromise vascular health. These cumulative disturbances of vascular homeostasis, coupled with impaired endothelial regeneration, significantly accelerate atherosclerosis development.

As periodontitis progresses, circulating inflammatory mediators such as CRP and IL-6 amplify systemic inflammation. CRP promotes platelet activation, immune modulation, and binding to oxLDL (oxidized low-density lipoprotein), promoting foam-cell formation and plaque progression. Cytokine-driven alterations in lipid metabolism, through hypothalamic-pituitary-adrenal signaling and macrophage gene reprogramming, increase cholesterol synthesis and exacerbate atherosclerosis. The IL-33/ST2 (suppression of tumorigenicity 2) axis also regulates vascular inflammation: CRP elevation increases sST2 (soluble ST2), which antagonizes IL-33 signaling and impairs its endothelial-protective effects, accelerating vascular injury. Elevated sST2 levels correlate with atherosclerotic burden and higher stroke risk (Figure 1).[44–46]

Apart from hematogenous translocation, oral microbes can colonize the gut when the oral-gut barrier is compromised, a frequent occurrence in aging or long-term proton pump inhibitor use.[47,48] In periodontitis, pathogens such as *P gingivalis* and *F nucleatum* migrate to the intestine, disrupting homeostasis.[49] *P gingivalis* increases permeability by downregulating tight junction proteins, while *F nucleatum* drives inflammation via cytokines (TNF-α and IL-6).[24] Furthermore, *P gingivalis* dysregulates immunity by expanding IL-9+ CD4+T-cells and promoting Th17 dominance via TLR4 signaling.[50] The hypothalamic-pituitary-adrenal axis modulates this system; its excessive activation releases inflammatory cytokines that impair gut-brain communication.[51] Compromised barriers allow lipopolysaccharides and immune cells to enter the CNS, triggering neuroinflammation, while gut signals further regulate microglial and astrocyte function.[52] Gut microbiota composition is critical in stroke pathophysiology. Ischemia disrupts microbial homeostasis, and preexisting dysbiosis, marked by opportunistic pathogens and reduced short-chain fatty acid producers, worsens poststroke survival.[53] Specific taxa correlate with severity: *Ruminococcaceae* are associated with poor outcomes, whereas *Enterobacter* enrichment correlates with milder cases.[52] Finally, direct oral-brain communication occurs via the trigeminal, olfactory, and facial nerves. *P gingivalis* and *T denticola* have been detected in brain tissue, linked to Alzheimer pathology and BBB disruption via gingipains and TLR signaling.[54] This confirms a multiaxis model where oral dysbiosis impacts CNS function through both direct neural pathways and indirect systemic/gut immunomodulation (Figure 2).[25,55]

## EPIDEMIOLOGICAL STUDIES

Several observational studies support the association between periodontitis and stroke (Tables 1 and 2). Fifteen original case-control studies examined the link between periodontitis and stroke, revealing high overall heterogeneity ($I^2$=71%). The combined results indicated a significant association, showing that individuals with periodontitis had approximately twice the risk of developing stroke compared with those without the condition (odds ratio, 2.32 [95% CI, 1.70–3.17]; $P$<0.00001).[56] The combined findings from the 22 cohort studies demonstrated significant heterogeneity ($I^2$=91%). The results were significant, indicating an association between periodontitis and stroke (relative risk, 1.22 [95 CI, 1.15–1.29]; $P$<0.00001).[56] The association between gingivitis and stroke is less clear. Six studies evaluating the link between gingivitis and stroke risk have shown high heterogeneity ($I^2$=95%) with significant variations in the odds ratio across studies. The overall findings from this meta-analysis indicate no significant association between gingivitis and stroke risk ($P$=0.78).[56] The differences observed across various studies might be due to the diverse populations examined. People who are predisposed to atherosclerosis and its associated risk factors are understandably more susceptible to gingivitis; however, it involves milder inflammation than periodontitis.[56] A recent systematic review and meta-analysis of 14 studies (n=35 937; age >17 years) examined the link between PD and stroke. Thirteen studies assessed periodontitis via clinical attachment loss, and 6 evaluated gingivitis via the gingival index, with 5 overlapping

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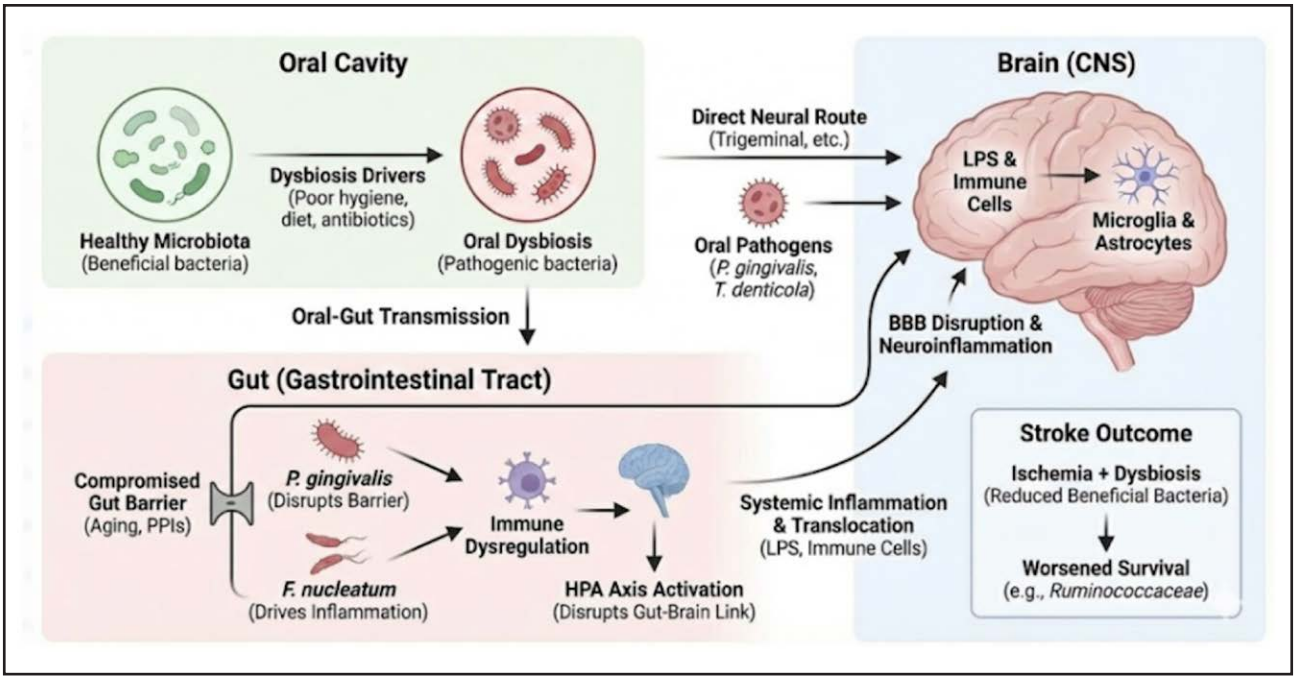


**Figure 2. Integrated mechanisms of the oral-gut-brain axis in stroke pathophysiology.**
This schematic illustrates the multipathway influence of oral dysbiosis on central nervous system (CNS) health and stroke outcomes. **Left**, Drivers such as poor oral hygiene, aging, and antibiotic use precipitate oral dysbiosis. Pathogens (eg, *P gingivalis* and *F nucleatum*) translocate to the gastrointestinal tract, particularly when the oral-gut barrier is compromised (eg, by proton pump inhibitors). **Bottom**, In the gut, *P gingivalis* increases permeability by disrupting tight junctions, while *F nucleatum* drives cytokine-mediated inflammation. This dysbiosis activates the hypothalamic-pituitary-adrenal (HPA) axis and facilitates the systemic translocation of lipopolysaccharides (LPS) and immune cells. **Right**, These inflammatory mediators and pathogens reach the CNS via systemic circulation, disrupting the blood-brain barrier (BBB), or through direct neural routes (eg, trigeminal nerve). Within the brain, they trigger neuroinflammation via microglial and astrocyte activation. Following a stroke, ischemia-induced gut dysbiosis (characterized by a reduction in beneficial bacteria) creates a negative feedback loop that exacerbates injury and worsens survival outcomes.

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both conditions. Meta-analysis revealed a significant association between periodontitis and stroke (effect size, 1.32 [95% CI, 1.04–1.60]) and a weaker, less precise association for gingivitis (effect size, 1.17 [95% CI, 0.42–1.92]).[57]

Several studies have established a significant association between PD and an increased risk of ischemic stroke. Sen et al[58] conducted a cohort study that demonstrated PD as an independent risk factor for ischemic stroke, with a graded relationship between the severity of PD and stroke risk. Importantly, the study found that gingival inflammation, even in the absence of severe periodontal damage, elevated the risk of ischemic stroke, highlighting the link of inflammation over tissue attachment loss in this association.[58] These findings align with a retrospective cohort study from Korea, which followed patients for over a decade and confirmed that PD was linked to a higher incidence of acute myocardial infarction, stroke, and major adverse cardiovascular events, with risks increasing by 11%, 3.5%, and 4.1%, respectively. Moreover, the Korean study found that age and sex modulated this association, with females and individuals aged 40 to 59 years exhibiting a stronger relationship between severe PD and cardiovascular

**Table 1. Inflammatory Cascades Linking Periodontal Pathogens to Vascular Inflammation**

| Cascade/model | Pathogen | Endothelial/vascular readout |
|---|---|---|
| TLR2/4→MyD88→NF-κB→IL-6/CRP | *P gingivalis* | ↓NO, ↑adhesion molecules; systemic cytokines |
| Atherosclerosis acceleration | *P gingivalis* | ↑Plaque burden in ApoE$^{-/-}$ mice |
| Junctional injury/permeability | *P gingivalis* (gingipains and OMVs) | VE-cadherin/PECAM-1 cleavage; ↑permeability |
| Endothelial activation (ICAM-1/VCAM-1) | *A actinomycetemcomitans* (LtxA) | Brain EC activation; cell-cycle effects |
| TLR4-biased NF-κB activation | *F nucleatum* | MyD88-dependent NF-κB signature |
| Clinical biomarker and FMD link | Patients with PD | ↑IL-6/CRP; ↓FMD |

CRP indicates C-reactive protein; EC, endothelial cell; FMD, flow-mediated dilation; ICAM-1, intercellular adhesion molecule-1; IL, interleukin; LtxA, leukotoxin A; NF-κB, nuclear factor-κB; OMV, outer membrane vesicle; PD, periodontal disease; PECAM-1, platelet endothelial cell adhesion molecule-1; TLR, toll-like receptor; VCAM-1, vascular cell adhesion molecule-1; and VE-cadherin, vascular endothelial cadherin.

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**Table 2. Summary of Key Studies on Periodontitis and Stroke Risk**

| Study/source | Design | Sample size/population | Stroke outcome | Effect estimate | Notes |
|---|---|---|---|---|---|
| Meta-analysis | Case control | 15 studies | Any stroke | OR, 2.32 (95% CI, 1.70–3.17) | High heterogeneity ($I^2$=71%); association only, confounding likely |
| Meta-analysis | Cohort | 22 studies | Any stroke | RR, 1.22 (95% CI, 1.15–1.29) | High heterogeneity ($I^2$=91%); observational, residual confounding |
| Dewan et al meta-analysis, 2024 | Mixed | 14 studies | Any stroke | Pooled effect size, 1.32 (95% CI, 1.04–1.60) | For periodontitis and stroke |
| Sen et al | Cohort | Large US cohort | Ischemic stroke | HR ↑ with PD severity | Independent risk factor; inflammation>attachment loss |
| Korean cohort | Retrospective | >10-y follow-up | Stroke, AMI, and MACE | Stroke risk ↑ 3.5% | Age/sex modified effect; stronger in women, age 40–59 y |
| Lafon et al | Case control | Patients with stroke vs controls | Ischemic stroke | Association with BOP and bone loss | Higher CRP in patients with stroke; supports the inflammation hypothesis |
| PAVE pilot | RCT | Periodontal therapy vs control | CV events (including stroke) | No significant difference | High loss to follow-up; causality not established |
| PREMIERS trial | RCT | Patients with ischemic stroke/TIA | Composite vascular outcome | No significant difference | Short follow-up; intensive PD treatment did not reduce events |

AMI indicates acute myocardial infarction; BOP, bleeding on probing; CRP, C-reactive protein; MACE, major adverse cardiovascular event; OR, odds ratio; PD, periodontal disease; RR, relative risk; and TIA, transient ischemic attack.

disease outcomes.[59] Furthermore, severe periodontitis was independently associated with a higher risk of atrial fibrillation, and mediation analyses indicated that atrial fibrillation may partially account for the observed association between PD and cardioembolic stroke. Notably, individuals receiving regular dental care or periodontal therapy demonstrated lower atrial fibrillation incidence and reduced inflammatory biomarker levels, suggesting that controlling oral inflammation may help prevent atrial remodeling and decrease the risk of cardioembolic events.[60–62]

Conversely, other studies have suggested that the relationship between PD and cardiovascular events may not be as robust as initially thought and have also raised questions regarding causation versus association. For instance, the PAVE pilot study, which aimed to assess whether periodontal treatment could reduce cardiovascular risk, found that cardiovascular adverse events occurred with similar frequency in both the periodontal therapy and community dental care groups, indicating no significant difference in event rates. While there was a nonsignificant trend toward fewer serious adverse events in the periodontal treatment group, the overall pattern of adverse events was comparable between the 2 groups over the 25-month study period. Notably, the study experienced considerable loss to follow-up, which may have influenced the results and limited the ability to draw definitive conclusions about the impact of periodontal therapy on cardiovascular outcomes.[43] The PREMIERS trial was a multicenter phase II randomized clinical trial evaluating intensive versus standard PD treatment in patients with ischemic stroke or high-risk transient ischemic attack. Patients assigned to the intensive arm received immediate, comprehensive, biologically targeted periodontal therapy with ongoing maintenance and local antibiotic support, whereas the standard arm received more conservative care, which included full-mouth supragingival scaling using hand instruments and ultrasonic scalers followed by supragingival polishing with abrasive dental polishing paste, with monitoring and deferred intensive therapy, unless disease progression was detected. PREMIERS found no significant superiority of intensive treatment over standard periodontal treatment in reducing composite vascular outcomes (stroke, myocardial infarction, or death). Although both treatment arms showed improvements in secondary outcomes such as diastolic blood pressure and HDL (high-density lipoprotein) cholesterol levels, no significant differences in cardiovascular events were observed, and no significant time-treatment interaction was observed with any of the secondary outcomes, suggesting that secondary-outcome measures did not change by time and treatment group. In addition, the study noted a dampened effect of intensive treatment on systemic inflammation markers such as hs-CRP and HbA1c, which may have been driven by the increased systemic inflammation associated with more frequent periodontal treatment. Despite the randomized design, limitations such as a high proportion of patients with severe PD (stage IV periodontal profile class, which is known to be less amenable to PD treatments), a smaller-than-expected sample size, and a relatively short follow-up period (12 months) may have influenced the trial's results, suggesting that further studies with larger cohorts and longer follow-up are needed to more conclusively determine the role of periodontal treatment in reducing cardiovascular risk.[60]

Lafon et al[63] conducted a case-control study that demonstrated an independent association between PD, particularly bleeding on probing and bone loss, and ischemic stroke. Their findings align with prior studies, highlighting bone loss as the primary odontological factor linked to ischemic stroke, with patients with stroke

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exhibiting significantly higher levels of CRP. This supports the hypothesis that periodontal inflammation may contribute to increased systemic inflammation, potentially driving atherogenesis and ischemic events.[63] Meta-analyses of prospective studies have further confirmed these associations, showing that PD is linked to an elevated risk of major adverse cardiovascular events, coronary heart disease, myocardial infarction, stroke, cardiac death, and all-cause mortality. Potential mechanisms include the direct invasion of cardiovascular endothelial cells by periodontal pathogens, systemic inflammation induced by oral pathogens, and the activation of oxidative stress pathways that impact myocardial and vascular function. Interestingly, the relationship between PD and cardiovascular outcomes appeared stronger in Western countries, possibly due to older age and more severe PD in these populations.[64]

Although these studies reinforce a consistent association between periodontitis and stroke, the high heterogeneity among studies and residual confounding from shared vascular risk factors (eg, smoking, diabetes, and socioeconomic status) limit causal inference. Differences in case definitions of periodontitis, stroke subtypes, and adjustment strategies likely contribute to variability across studies. Future prospective and mechanistic investigations that integrate standardized periodontal assessments with biomarker and imaging correlations are needed to clarify whether periodontitis acts as an independent modifiable risk factor or primarily reflects systemic inflammatory burden.

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## CLINICAL IMPLICATIONS

The PREMIERS trial evaluated whether intensive PD treatment could reduce the risk of recurrent vascular events (stroke, myocardial infarction, or death) compared with standard treatment. Although intensive treatment was not superior to standard treatment with respect to either the primary or secondary outcomes, its results suggest that both standard and intensive PD therapy may offer improvements in cardiovascular risk factors, such as blood pressure and lipid profiles.[65] Beyond PREMIERS, other interventional strategies, including dental scaling, subgingival curettage, root planing, adjunctive antibiotic therapy, and structured oral hygiene programs, have been tested as secondary prevention measures for stroke and cardiovascular disease. Evidence from a 10-year retrospective cohort study indicates that both dental scaling and intensive periodontal interventions were associated with a reduced risk of subsequent ischemic stroke.[66] Similarly, a Taiwanese study demonstrated that periodontal treatment and dental prophylaxis significantly lowered stroke risk, particularly among younger adults.[67] These findings underscore the potential benefit of maintaining periodontal health as part of vascular risk reduction.

However, in randomized trials such as PREMIERS and PAVE that directly compared intensive and standard periodontal treatment strategies, intensive therapy was not associated with a reduction in major cardiovascular events. Several factors may explain these neutral results: short follow-up durations, small sample sizes, high baseline severity of PD, and potential inflammatory rebound following intensive therapy. For example, frequent periodontal interventions can transiently increase systemic inflammation, potentially offsetting long-term benefits. In addition, loss to follow-up and underpowered designs limit the ability to detect clinically meaningful differences. These limitations highlight the need for larger, longer-term studies to clarify whether periodontal treatment can influence hard vascular outcomes.

From a preventive care perspective, oral health screening may serve as a valuable component of vascular risk assessment. PD shares common risk factors with atherosclerosis, such as smoking, diabetes, and obesity, and contributes to systemic inflammation, a recognized driver of vascular pathology. Integrating periodontal evaluation into cardiovascular prevention strategies aligns with current American Heart Association/American Stroke Association guidelines, emphasizing comprehensive risk factor management. While formal recommendations for routine periodontal screening are not yet established, emerging evidence supports its inclusion in multidisciplinary approaches to stroke prevention.

Collectively, these studies highlight the potential role of oral health in reducing stroke risk and reinforce the importance of integrated care models that address PD alongside traditional cardiovascular risk factors.

## FUTURE DIRECTIONS

To advance our understanding of the specific mechanisms linking cerebrovascular disease to PD, it is crucial to focus on several key areas. First, separating cerebrovascular disease cases not only into ischemic and hemorrhagic categories but also into etiological subcategories (intracranial atherosclerotic disease, carotid artery disease, cardioembolic strokes, small vessel disease, intracranial hemorrhage, brain aneurysms, and so on) will provide more precise insights. In addition, standardizing reporting by adjusting for cardiovascular risk factors will enhance the reliability of findings. Conducting prospective studies with robust statistical modeling is essential to establish clear causal relationships. The PREMIERS Intracranial Study is currently exploring the impact of PD treatment in stroke survivors with moderately severe periodontitis on both cerebral small vessel disease progression, as assessed by FLAIR imaging, and poststroke cognitive decline, measured by standardized cognitive assessments. The trial will also examine whether the

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effects of periodontal treatment differ between Black and White patients and explore whether changes in systemic inflammatory markers are associated with the rate of CSVD progression and cognitive decline after stroke.[68] These ongoing efforts underscore the importance of addressing periodontal health to potentially mitigate cerebrovascular disease risk.[69]

## CONCLUSIONS

This review underscores the significant impact of dental diseases on stroke risk, highlighting the complex interplay between periodontal pathology, bacteremia, and systemic inflammation. The evidence suggests that dental diseases contribute to multiple vascular pathways that increase stroke risk, particularly through mechanisms such as bacterial dissemination and inflammatory responses that exacerbate endothelial dysfunction. Studies have shown that effective management of periodontal health, including regular dental cleanings and more intensive treatments such as scaling and root planning, may improve cerebrovascular risk factors. These findings advocate for the integration of oral health care into broader cardiovascular risk management strategies, emphasizing the potential of preventive dental care as one of the components in reducing stroke incidence.

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## ARTICLE INFORMATION

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